

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 27)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

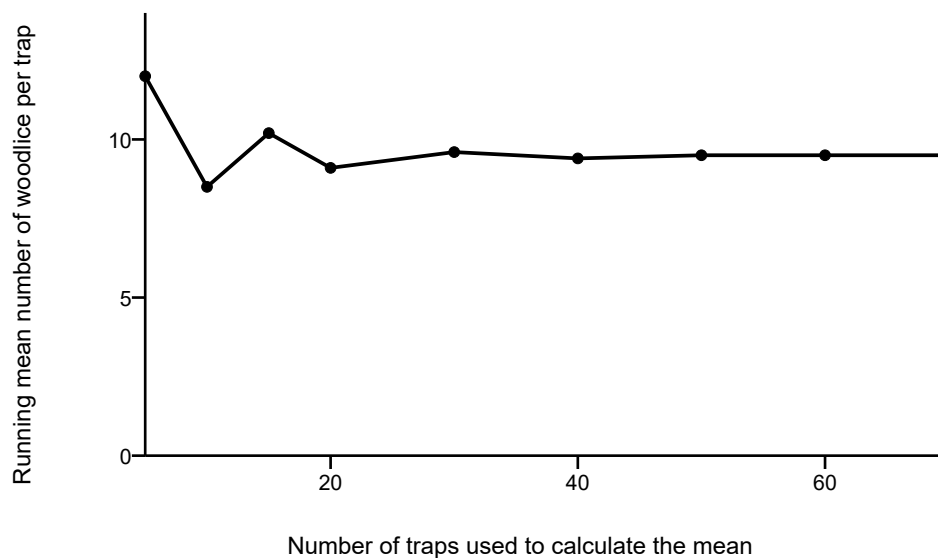
Question	Mark
1	
2	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 1 . 1

Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

0 1 . 2

Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

0 1 . 3

The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

0 1 . 4

Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

10

0 2 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

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0	2	.	2
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Explain why growing coral reefs help to resist climate change.

[2 marks]

0	2	.	3
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Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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0 3 . 1

Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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2

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0 3 . 2

Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 3 . 3

Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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5

0 4

The Isle of Eigg (area 31 km², population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 4 . 1

Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

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0 4 . 2

Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

1

2

3

0 4 . 3

Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

0 4 . 4

Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

10

05 . 1

Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

- A Its mass ☐
- B Its magnetic field ☐
- C Its distance from the Sun ☐
- D Its speed of rotation ☐

05 . 2

Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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05 . 3

Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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06 . 1

Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species

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Endemic species

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Keystone species

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Flagship species

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06 . 2

Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

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07 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

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07 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

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07 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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5

08 . 1

Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

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08 . 2

Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

08 . 3

Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

08 . 4

Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

10

09

Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha ⁻¹	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

09**. 1**

Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

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09**. 2**

Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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09 . 3

Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

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10

Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

10 . 1

Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer

 %

10 . 2

Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

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10 . 3

Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

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10 . 4

Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

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Write an essay on one of the following topics.

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Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

1 1 . 2

Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

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END OF QUESTIONS